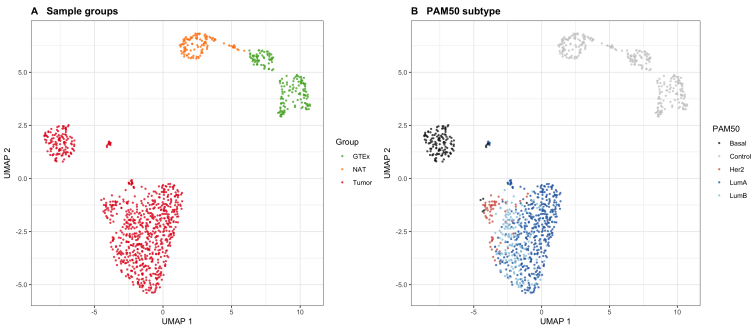
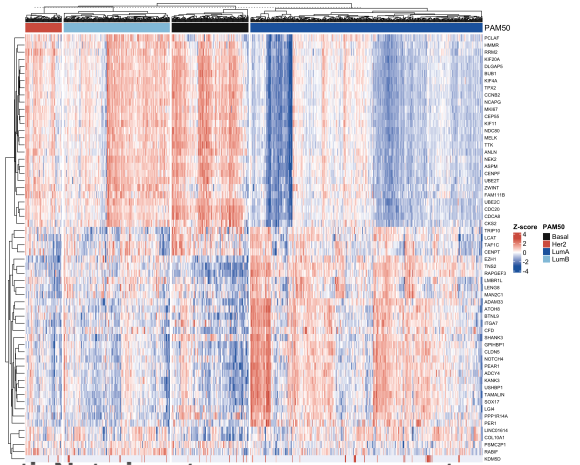
# TCGA–BRCA PAM50 Subtype Analysis – Pipeline Summary

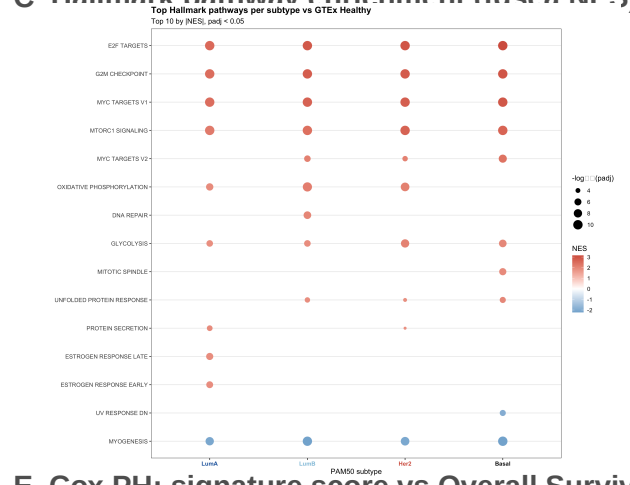
## A UMAP – sample groups and PAM50 subtypes



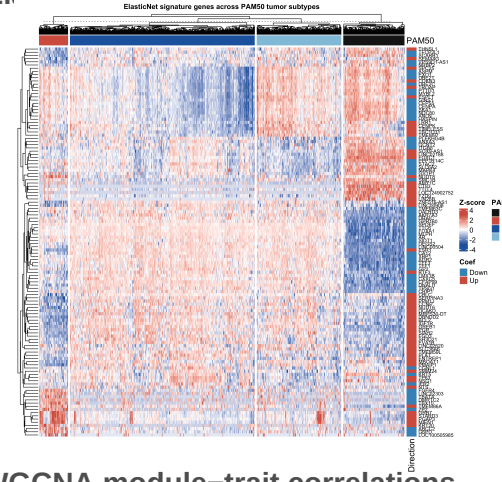

## B Top DEGs per subtype vs GTEx (z-scored)



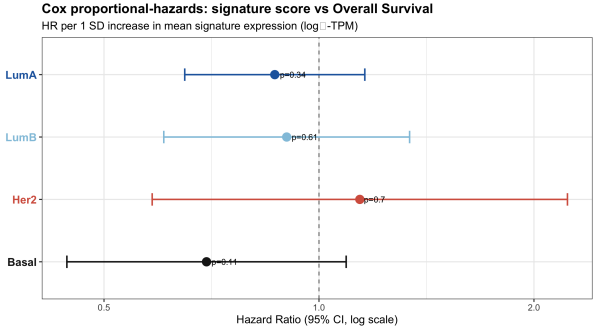

## C Hallmark pathway enrichment (fgsea NES)

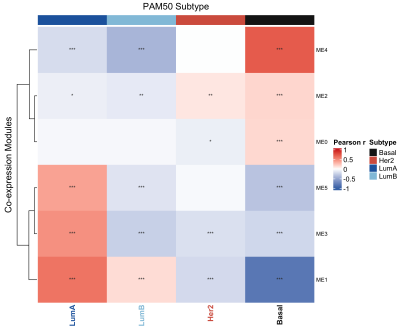



## D ElasticNet signature genes across tumors




## E Cox PH: signature score vs Overall Survival




## F WGCNA module–trait correlations